Superstore Retail Performance and State-Level Socioeconomic Context

ETW2001 Assignment 2

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Page 1 — Dashboard Screenshot

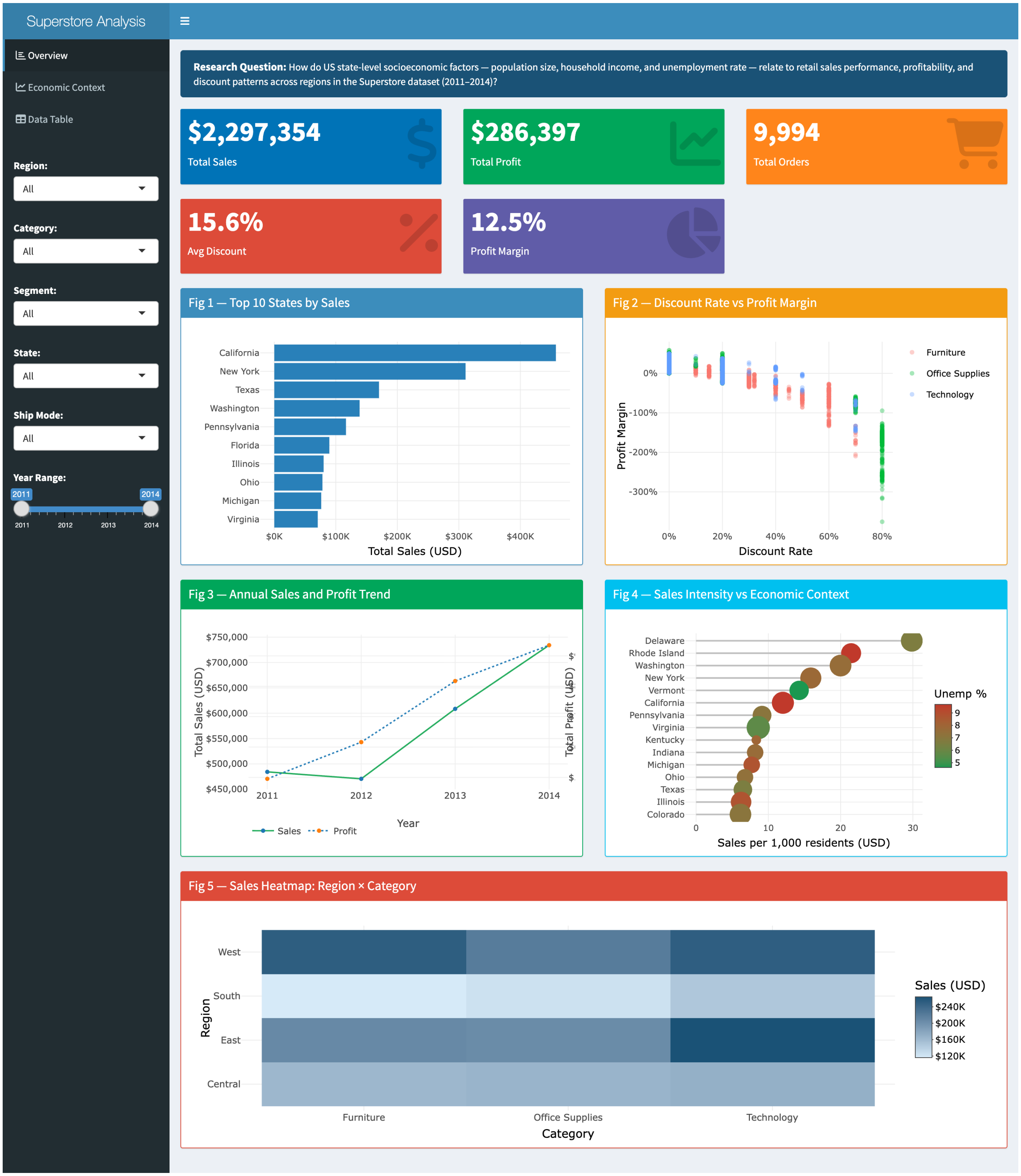


Figure 1: R Shiny dashboard showing three interactive tabs — Overview (five core charts, six sidebar filters), Economic Context (three external-dataset charts), and Data Table. All visualisations update dynamically with filter changes.

Page 2 — Problem Design

Research Question: *How do US state-level socioeconomic factors — population size, median household income, and unemployment rate — relate to retail sales performance, profit margins, and discount patterns across regions in the Superstore dataset (2011–2014)?*

Retail performance is influenced by both internal operations and external market conditions. States with larger populations, stronger consumer incomes, and lower unemployment are expected to generate higher demand and support healthier margins. This analysis contextualises Superstore transaction data with three authoritative external datasets to explore whether observable sales and profitability patterns are consistent with state-level economic characteristics.

The study covers the 49 US jurisdictions represented in the Superstore data — the 48 contiguous states plus the District of Columbia — across 2011–2014. Each external indicator is drawn from the **same 2011–2014 window** and joined to the Superstore transactions on **State and Year**, so the economic context is matched to the analysis period rather than approximated from a later one. All associations are descriptive; no causal claims are made.

Scope: Five visualisations address sales concentration, discount behaviour, time trends, category mix, and regional patterns. An Economic Context tab in the dashboard links Superstore performance directly to the three external datasets.

Page 3 — Dataset Selection

Superstore (primary). 9,994 US retail orders, 2011–2014, from Assignment 1 (Sales, Profit, Discount, Category, Region, State, Order Date). Three external datasets supply the state economic context the transactions lack, each matched to the 2011–2014 window and joined on State and Year:

Table 1: External datasets, period-matched to 2011-2014 and joined on State and Year.

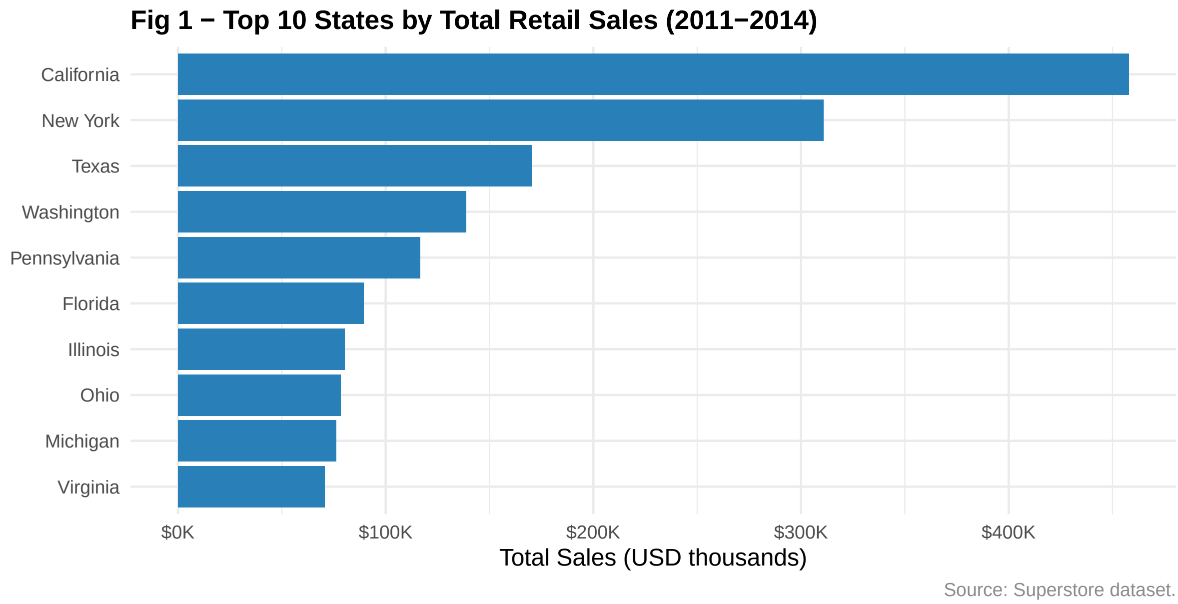
Dataset	Source	Years	Unit	Join Key	Justification
Resident population by state	US Census Bureau (PEP)	2011-2014	Persons	State + Year	Market-size denominator: lets sales be read per capita, not only in absolute terms.
Median household income by state	US Census Bureau (SAIPE)	2011-2014	USD (current)	State + Year	Proxies household purchasing power against which sales and margins can be assessed.
State unemployment rate	US BLS (LAUS)	2011-2014	% labour force	State + Year	Captures post-recession economic stress during 2011-2014 for comparison with margins.

All three are authoritative US-government series, downloaded as key-free public files and rebuilt reproducibly by `scripts/prepare_external_data.R` (full provenance in `data/dataset_provenance.csv`). They complement Superstore by supplying the *demand context* — how many people, how prosperous, and how economically stressed each state was — that the transaction records alone do not contain.

References

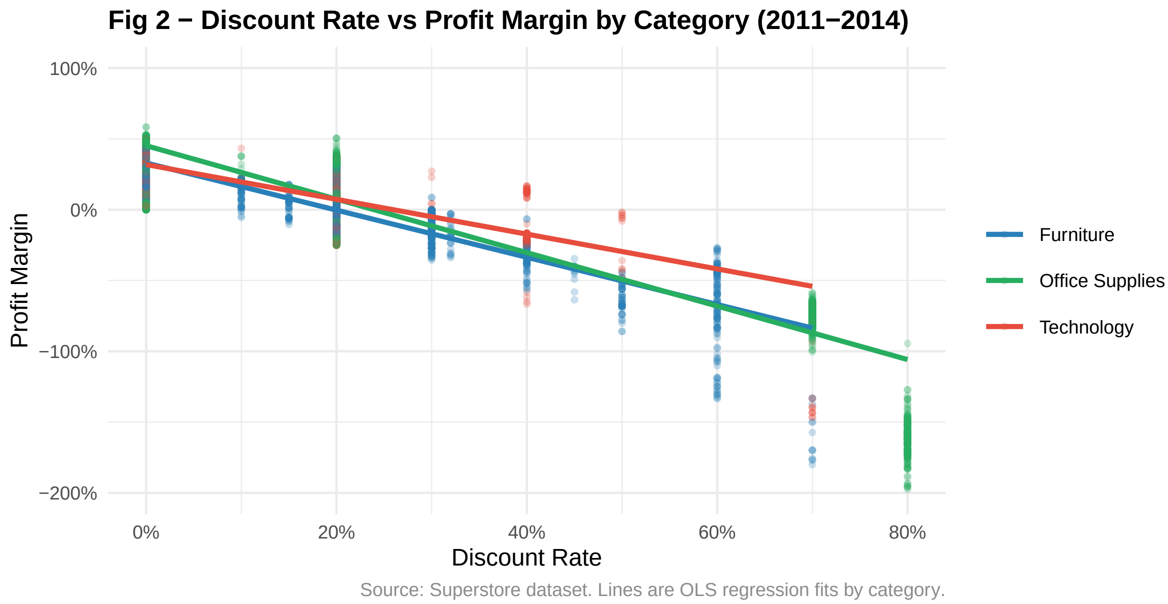
US Census Bureau, Population Division (2019). *Annual Estimates of the Resident Population for the United States, Regions, and States: 2010–2019* (NST-EST2019-alldata). https://www2.census.gov/programs-surveys/popest/datasets/2010-2019/national/totals/nst-est2019-alldata.csv (accessed 30 May 2026). US Census Bureau (2011–2014). *Small Area Income and Poverty Estimates (SAIPE): state median household income*. https://www.census.gov/programs-surveys/saipe/data/datasets.html (accessed 30 May 2026). US Bureau of Labor Statistics (2011–2014). *Local Area Unemployment Statistics (LAUS): statewide unemployment rates*. https://www.bls.gov/lau/ (accessed 30 May 2026).

Page 4 — Visualisation 1: Top 10 States by Sales



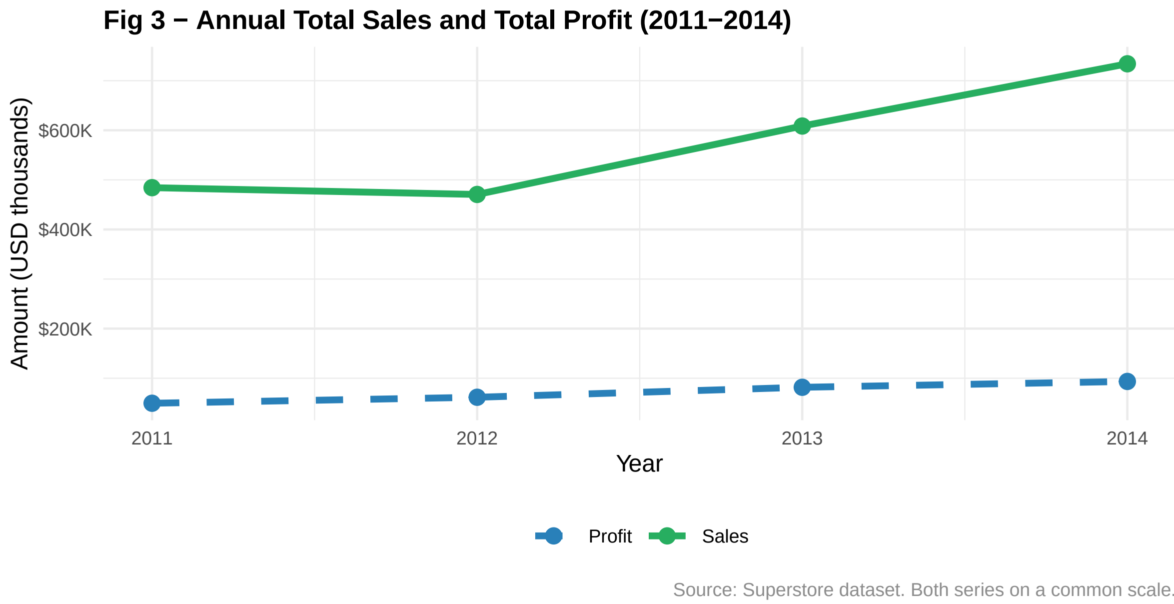
Interpretation. California, New York, and Texas account for the largest share of total sales — a pattern consistent with their status as the three most populous US states (US Census Bureau PEP, 2011–2014). Across all 49 states, state population and total sales are strongly correlated (r $\approx$ 0.90; see Figure 4 and the dashboard's Economic Context tab), so state population is the variable most strongly associated with retail volume. Pennsylvania and Florida outperform their population rank somewhat, possibly reflecting stronger logistics infrastructure or concentrated corporate buyer activity. For resource allocation, weighting operational investment toward large-population states is likely to maximise absolute return, though the Economic Context charts in the dashboard reveal that absolute sales do not guarantee proportionally higher margins.

Page 5 — Visualisation 2: Discount Rate vs Profit Margin



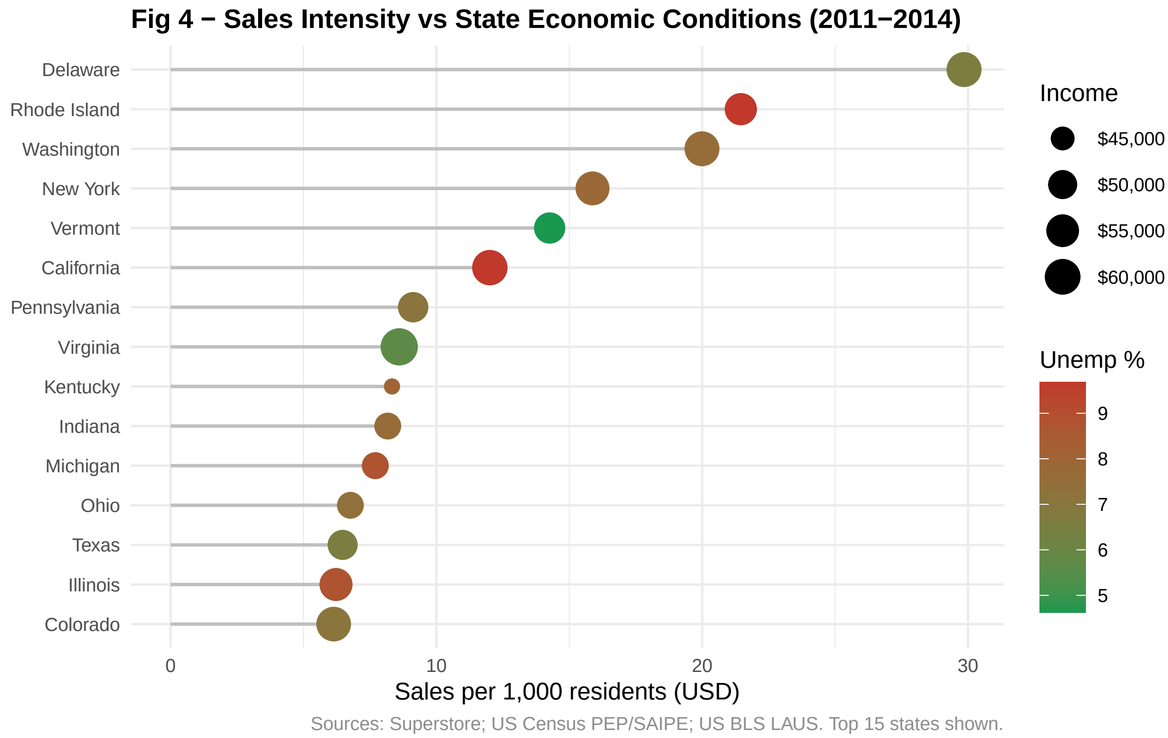
Interpretation. A consistent negative association between discount rate and profit margin is visible across all three product categories. Furniture shows the steepest decline, with many transactions above 30% discount producing negative margins. Technology maintains positive margins at moderate discount levels, consistent with the higher unit prices and stronger brand loyalty typical of technology goods. Office Supplies occupies an intermediate position. This pattern is consistent with the hypothesis that deep discounting erodes value at scale, and does not establish causality — margin outcomes also reflect cost structure and order volume. The finding suggests that category-specific discount guardrails, particularly for Furniture, may protect profitability without meaningfully reducing customer acquisition.

Page 6 — Visualisation 3: Annual Sales and Profit Trend



Interpretation. Total sales grew steadily from 2011 to 2014, reflecting an expanding retail business. Profit followed a broadly similar trajectory but with greater variability, and profit growth lagged behind sales growth in certain years — consistent with increased discount activity or a shift toward lower-margin product mix. The converging gap between the two series in 2014 is an encouraging signal of improving profitability discipline. Overall profit margin across the full period is approximately 12%, indicating meaningful room for improvement, particularly if high-discount behaviours identified in Figure 2 are addressed.

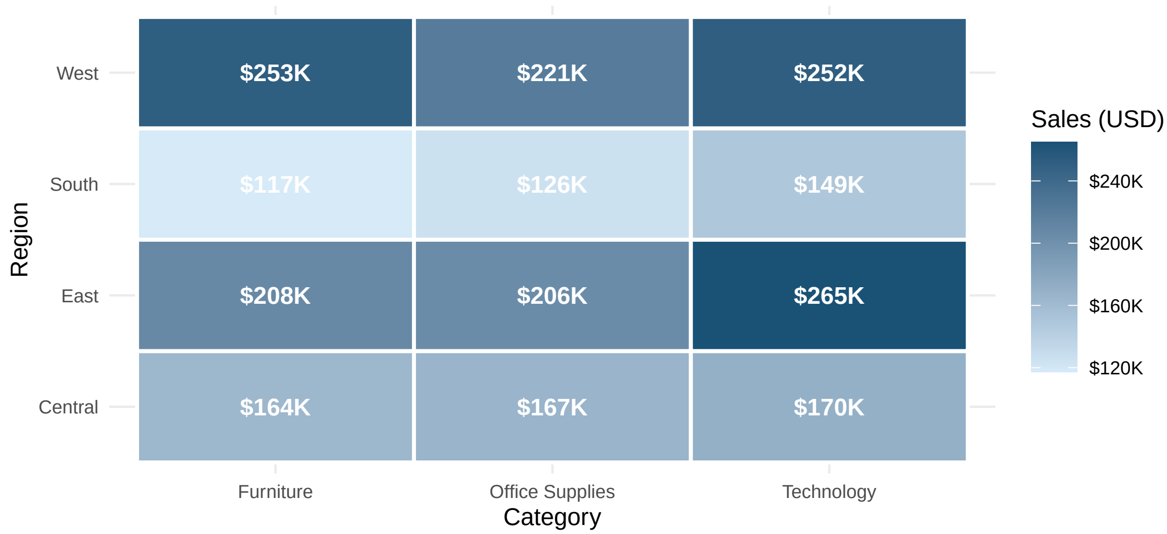
Page 7 — Visualisation 4: Sales Intensity vs State Economic Conditions



Interpretation. This chart is the report's direct test of whether state economic conditions track retail performance, and it combines all three external datasets at once: sales per 1,000 residents on the axis (population), colour for unemployment, and point size for median income. Sales intensity is highest in *small*-population states — Delaware, Rhode Island, Vermont — while the three largest absolute markets (California, New York, and Texas from Figure 1) fall to mid-table once population is divided out. Crucially, neither colour nor size forms a gradient down the ranking: low-unemployment Vermont (4.6%) and high-unemployment Rhode Island and California (9.7%) sit near the top together, and median income barely varies among the leaders. Across all 49 states these associations are weak (sales-per-capita with income r $\approx$ 0.23, with unemployment r $\approx$ 0.22), whereas absolute population explains absolute sales far better (r $\approx$ 0.90). The message is clear: *market size*, not local prosperity or labour-market health, is what tracks where Superstore sells — an insight visible only by joining the external data to the transactions.

Page 8 — Visualisation 5: Sales Heatmap — Region × Category

Fig 5 − Sales Heatmap by Region and Category (2011−2014)



Source: Superstore dataset.

Interpretation. The West and East regions generate the highest total sales across all three categories, driven by the two largest absolute markets — California (West) and New York (East), the top two states in Figure 1. This regional concentration tracks population and market size rather than local prosperity: California's median household income ranks only 9th and New York's 15th of 49 (the highest-income states are Maryland, New Jersey, and Connecticut), and Figure 4 shows income and unemployment are only weakly associated with sales. The South's relative strength in Technology is notable and may reflect faster technology adoption in southern metropolitan areas. Central-region underperformance across all categories suggests potential for strategic expansion, particularly in Office Supplies where logistics-efficiency advantages may lower the cost of market entry.

Page 9 — Data Story (Part 1)

The Geography of Retail Performance: Population, Prosperity, and Profitability

This analysis asked whether state-level socioeconomic conditions are associated with Superstore retail performance. Across five visualisations and three external datasets, a coherent and actionable picture emerges.

Population tracks volume — but prosperity does not. Figure 1 shows sales concentrated in California, New York, and Texas, the three most populous US states, and the Population vs Sales bubble chart in the Economic Context tab shows a strong positive association ($r \approx 0.90$). More residents means a larger customer base. Yet population is almost the *only* economic variable that tracks performance. Figure 4 shows that once sales are expressed per capita, neither median household income ($r \approx 0.23$) nor unemployment ($r \approx 0.22$) aligns with sales intensity, and income's association with margin is negligible ($r \approx 0.12$). High-income states are not the heaviest buyers, and economically stressed states are not the lightest — Superstore's footprint follows headcount, not prosperity.

Discounting is the primary profitability lever. Figure 2 presents the most operationally important finding: discount rate and profit margin are negatively associated across all product categories, most sharply in Furniture. This pattern is visible regardless of region or time period. Given that total sales grew consistently (Figure 3), the moderate overall profit margin (about 12%) is likely explained, at least in part, by widespread discount behaviour. Addressing discount depth in Furniture — where negative-margin transactions are common above 30% — represents the clearest near-term opportunity to improve returns.

Regional concentration creates both strength and vulnerability. Figure 5 shows West and East regions dominating sales across all categories. This concentration reflects the *populations* of California, New York, and Washington far more than their prosperity — California's income ranks only 9th of 49. Because volume rests on a handful of large-population states, the business is disproportionately exposed to economic or competitive shifts in those markets. The South's relative Technology strength (Figure 5) and the Central region's lower saturation suggest expansion opportunities in underserved markets.

Page 10 — Data Story (Part 2)

Limitations. Four caveats apply. (1) *Ecological inference* — the external indicators are state aggregates joined to state retail totals, so associations describe states, not individual customers or orders. (2) *Per-capita artifact* — sales per 1,000 residents is inflated for small-population states and reflects Superstore's sampled footprint, which is not proportional to population; the per-capita leaders in Figure 4 should not be read as the "best" markets. (3) *Source nature* — SAIPE income is a modelled estimate and unemployment is a coincident indicator, so both contextualise rather than measure customer behaviour. (4) *Observational design* — margins also depend on cost structure, product mix, and discount policy that state economics does not capture, so no causal claims are made.

Recommendations. Three evidence-grounded recommendations follow:

1. *Furniture discount guardrails* — Figure 2 shows discounts above 20% are consistently associated with negative Furniture margins. Capping approved discounts in this category is the clearest near-term lever to protect profit without materially reducing conversion.
2. *Do not equate large markets with healthy ones* — population predicts volume ($r \approx 0.90$) but not profitability: 10 of the 49 states are loss-making, including Texas (margin −15%) despite low unemployment (6.5%). High-volume states should be audited for discount and margin discipline (Figure 2, Economic Context tab) rather than assumed healthy because they are large.
3. *Size territories on population, not prosperity proxies* — because population correlates strongly with sales ($r \approx 0.90$) while income and unemployment correlate weakly ($r \approx 0.2$; Figure 4), state population is the more defensible input for territory sizing and quota allocation; local income and unemployment should be treated as minor adjustments, not primary drivers.

Page 11 — AI Acknowledgement and Citations

AI Usage Acknowledgement. Claude (Anthropic) was used to assist with R Shiny layout coding, ggplot2 chart generation, and drafting initial paragraph outlines. All analytical interpretations, dataset selections, and conclusions were reviewed and verified by the author. The author accepts full responsibility for all submitted content. No sources were fabricated. AI-generated code was tested before inclusion.

Full References

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